Ranveer Deshmukh
22108A0061
EXCS 'A'

REM
Assignment - 1
National Energy Scenario

India's energy sector is one of the fastest growing in the world, driven by rapid industrialization, urbanization, and population growth. As the third-largest energy consumer globally, India faces the dual challenge of meeting rising energy demand while ensuring sustainability and reducing environmental impact.

- Current energy Mix

India's energy mix is still dominated by fossil fuels, particularly coal, which accounts for nearly 50-55% of total power generation. Oil and natural gas also play significant roles, especially in transportation and industrial sectors.

Growth in Renewable Energy:

In recent years, India has made remarkable progress in renewable energy deployment. The country has set an ambitious target of 500 GW of non-fossil fuel capacity by 2030. Solar energy leads the transition, while wind energy also contributes significantly.

Challenges

- Dependence on fossil fuels
- Energy security concerns
- Infrastructure limitations
- Financial stress

Government Initiatives

- National Solar Mission
- UJALA Scheme
- Perform, Achieve and Trade (PAT)
- Green Hydrogen Mission